EE_299 Project Presentation

FPGA Implementation of Chirp-Coded Subharmonic Ultrasound Imaging with Volterra Filtering

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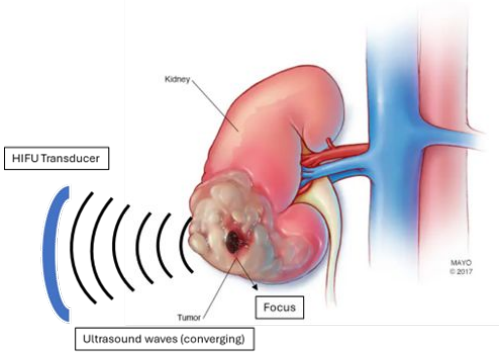
Ultrasound for therapy

[1]



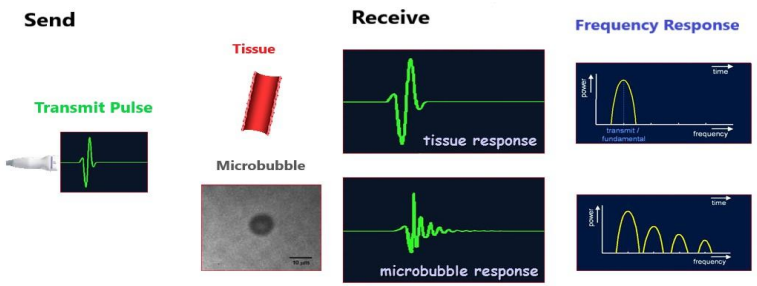
- Ultrasound is well known diagnostic imaging modality due to its non-invasive, non-ionizing and inexpensive nature.
- It is used in therapeutic applications such as targeted tumor ablation, improvement in drug delivery, blood-brain barrier opening etc.
- Monitoring of therapy is essential to track treatment progression and avoid damage to the tissues.

[2]

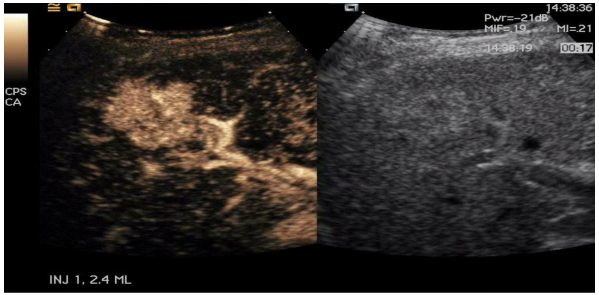
1. https://www.mayoclinic.org/diseases-conditions/kidney-cancer/symptoms-causes/syc-20352664
2. https://utswmed.org/medblog/pregnancy-ultrasound-scary/

Contrast Enhanced Ultrasound Imaging(CEUS)

- **Used Microbubbles:** uses microbubble agents to detect microvascular blood flow.
- **Nonlinear Signals:** Microbubbles oscillate nonlinearly, producing unique harmonic and subharmonic signals that tissue does not.
- **High Contrast:** Subharmonic imaging (at $f_0/2$) provides a high contrast-to-background ratio (CTR) by isolating these bubble-specific signals.



[1]



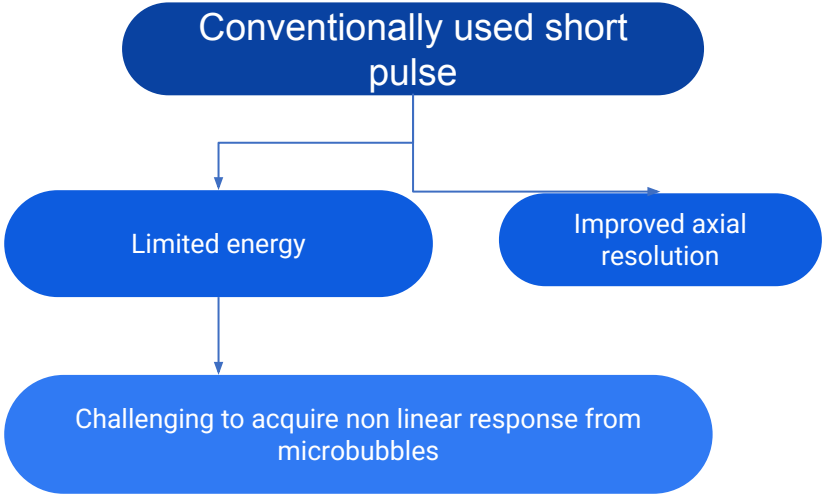
[2]

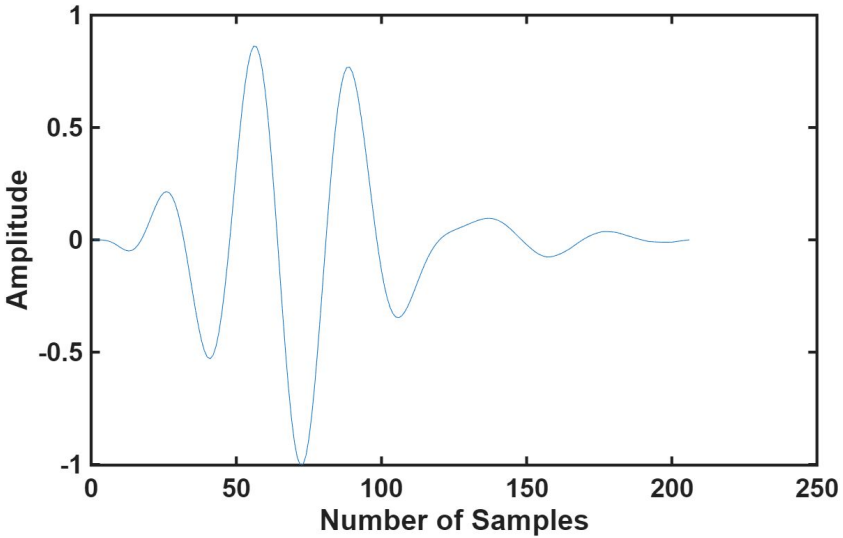
CEUS B-mode Image

Conventional B-mode image

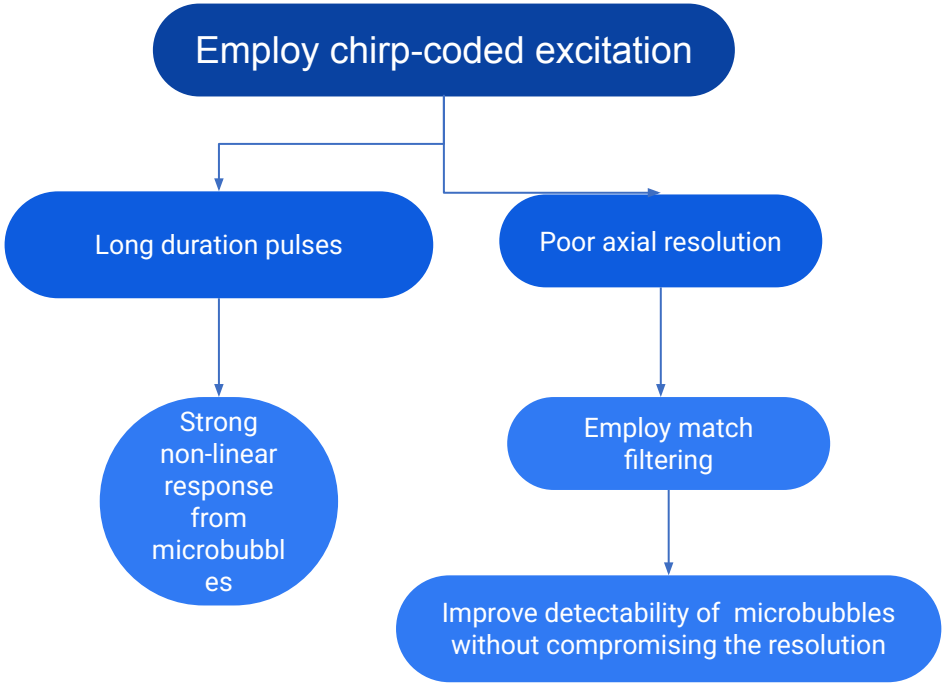
1.https://www.contrastultrasound-modality.com/contrast-ultrasound/what-microbubble/microbubbles-non-linear-oscillation
2.Burns P, Bubbles and how ultrasound systems talk to them: In: Contrast-Enhanced Ultrasound (2 Ed), Uni-Med press Bremen, 2011, pp.16-26.

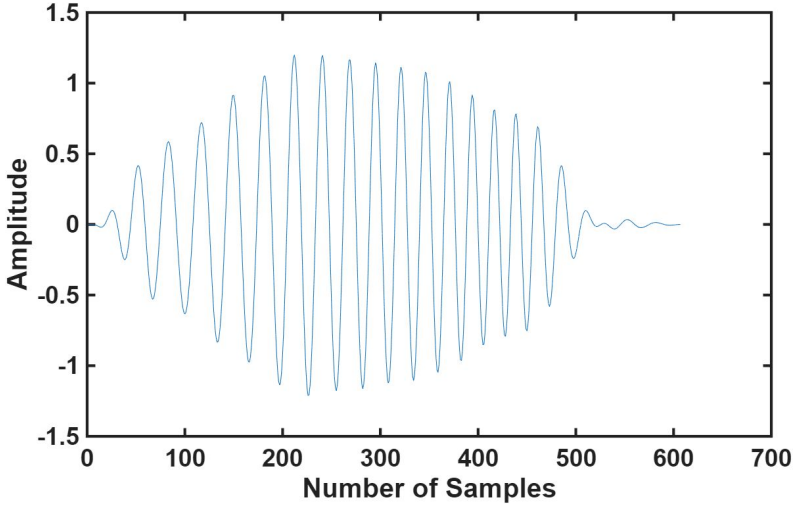
Contrast Enhanced Ultrasound Imaging with Conventional Ultrasound Pulse



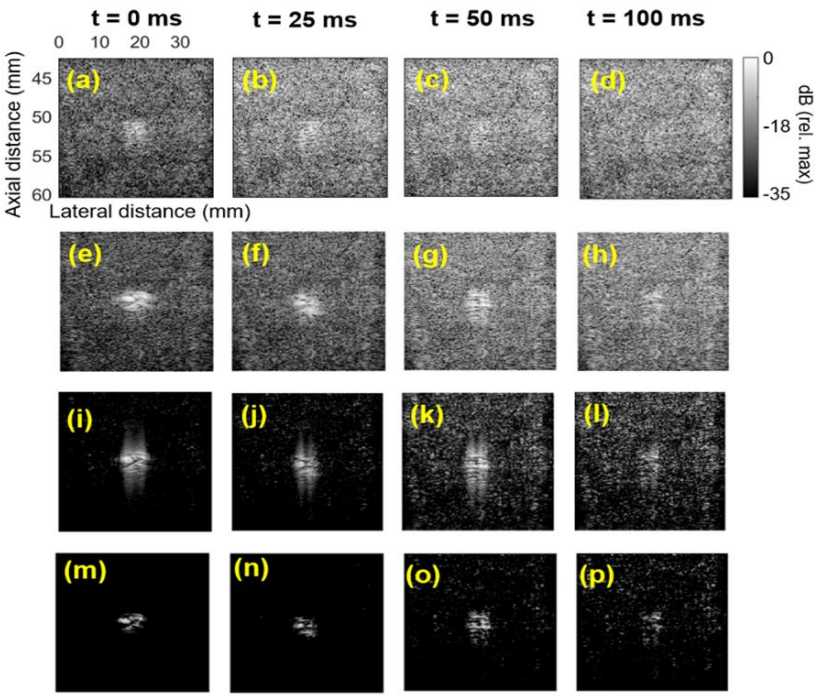


Contrast Enhanced Ultrasound Imaging with Chirp Pulse





Chirp-coded excitation with Volterra Filtering

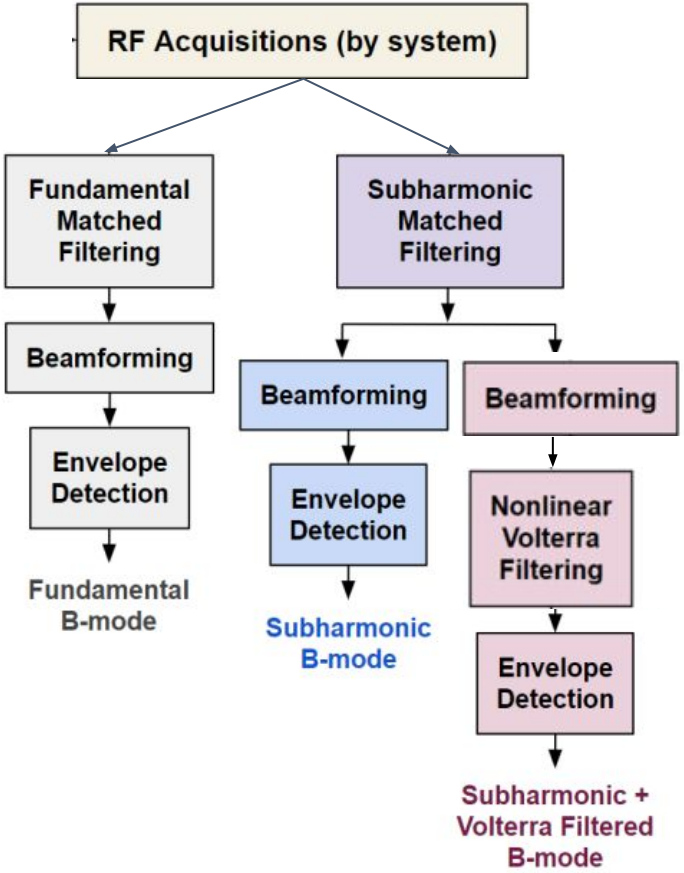


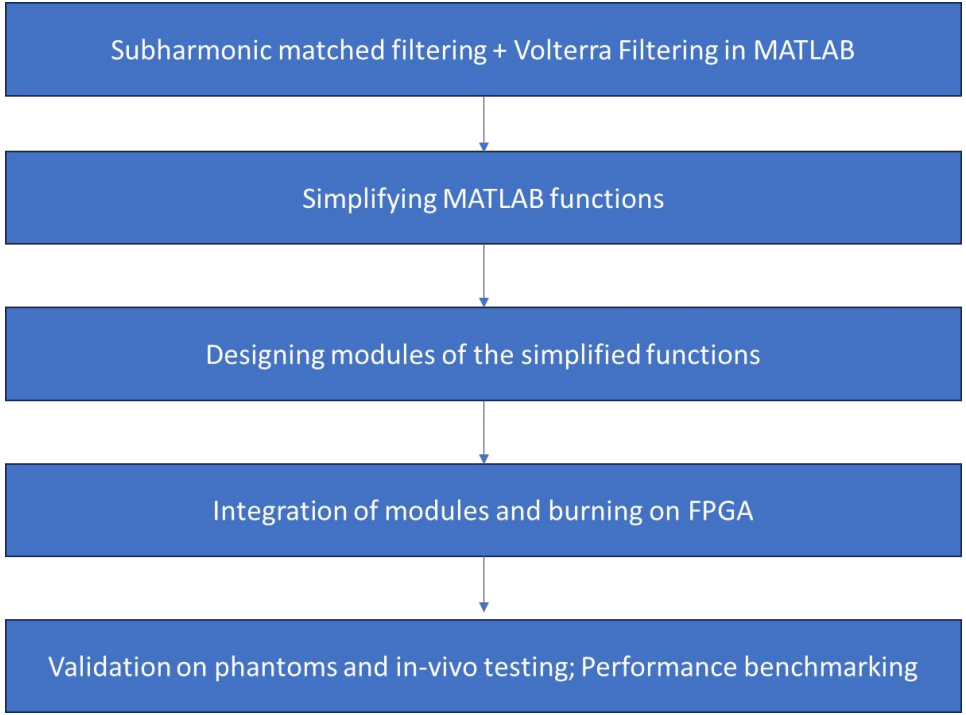
- Uses a high-energy, frequency-modulated (chirp) pulse to enhance the nonlinear response from the bubble cloud.

- A nonlinear Volterra filter is then applied to model and subtract the interfering signals that are generated by the surrounding tissue.

$$y[n] = \sum_{i=0}^{m-1} x[n-i]h_l[i] + \sum_{i=0}^{m-1}\sum_{j=0}^{m-1} x[n-i]x[n-j]h_q[i,j]$$
$$+ \sum_{i=0}^{m-1}\sum_{j=0}^{m-1}\sum_{k=0}^{m-1} x[n-i]x[n-j]x[n-k]h_c[i,j,k]$$
$$= y_l[n] + y_q[n] + y_c[n]$$

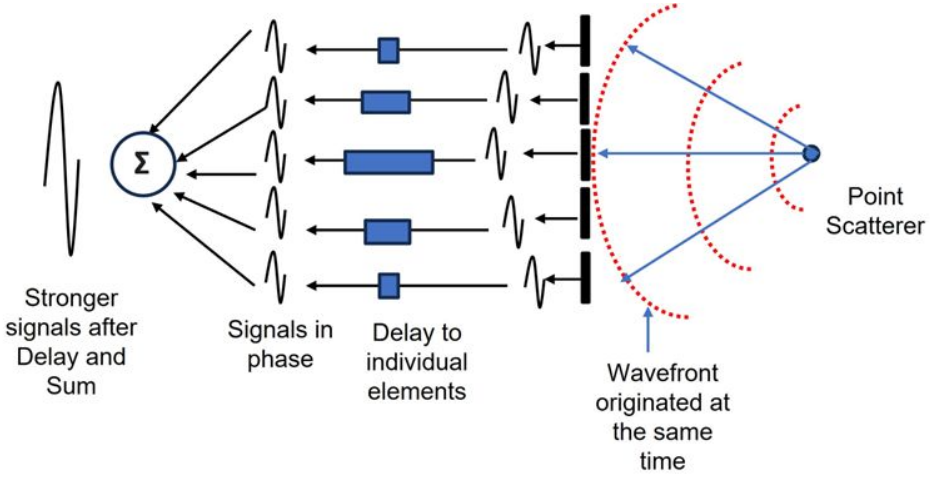
Pipeline


RF Acquisitions (by system)
Fundamental Matched Filtering
Subharmonic Matched Filtering
Beamforming
Beamforming
Beamforming
Envelope Detection
Envelope Detection
Nonlinear Volterra Filtering
Envelope Detection
Fundamental B-mode
Subharmonic B-mode
Subharmonic + Volterra Filtered B-mode

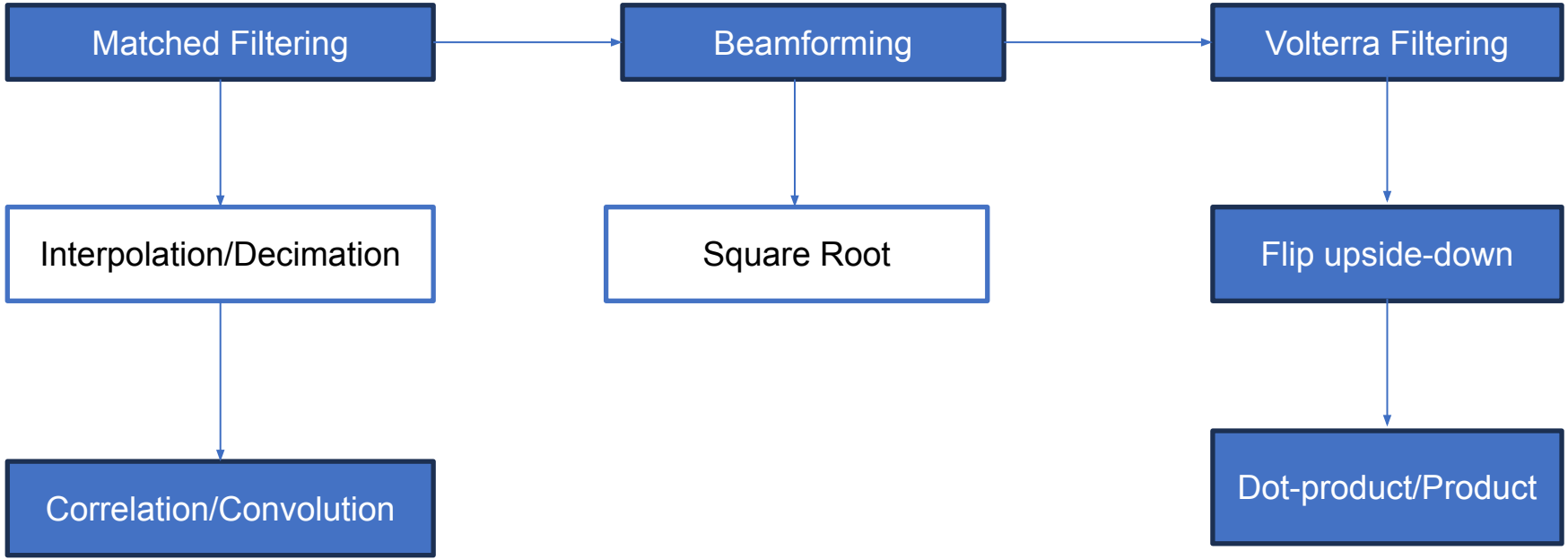

Subharmonic matched filtering + Volterra Filtering in MATLAB
Simplifying MATLAB functions
Designing modules of the simplified functions
Integration of modules and burning on FPGA
Validation on phantoms and in-vivo testing; Performance benchmarking

Beamforming

- Computation of delay map for the 128 transducers each of size 973x128.
- Total delay map size 973x128x128x(computation bits).
- Delay calculated for an individual pixel at an instant. Maximum vector size for a pixel would be 1x51 (keeping 25 neighbour elements)
- Lowers the FPGA's on chip-storage requirements.



Functions


Matched Filtering
Beamforming
Volterra Filtering
Interpolation/Decimation
Square Root
Flip upside-down
Correlation/Convolution
Dot-product/Product

Thank You!